

Banking Transaction Monitoring System

1. Introduction

The Banking Transaction Monitoring System is a web-based application developed to monitor financial transactions and detect suspicious activities. In today's digital banking environment, large volumes of transactions occur every second, making manual monitoring inefficient and error-prone.

This system provides an automated solution that helps identify fraudulent transactions using predefined rules. It ensures better security, transparency, and efficiency in handling banking operations. The application is built using Angular and TypeScript, offering a user-friendly interface and real-time transaction monitoring.

2. Problem Statement

Traditional banking systems often rely on manual verification or basic tools to monitor transactions. This leads to several issues such as delayed fraud detection, lack of real-time monitoring, and difficulty in analyzing large datasets.

Users may face risks like unauthorized transactions, while administrators struggle to track and verify transaction activities effectively. Therefore, there is a need for a system that can automatically monitor transactions, detect fraud, and provide a centralized platform for both users and administrators.

3. Proposed Solution

The proposed system is a Banking Transaction Monitoring System that provides role-based access and automated fraud detection.

The system includes two main modules:

- **User Module:** Allows users to log in and view their personal transaction history, including details like reference number, date, amount, type, and status. Users can also search and filter their transactions.
- **Admin Module:** Enables administrators to view all transactions, monitor fraud activities, filter user data, and update fraud detection rules dynamically.

The system uses rule-based logic such as high transaction amounts and unusual patterns to classify transactions as "Fraud" or "Normal".

4. System Features

- Secure login system with role-based access
- Transaction dashboard with real-time data
- Fraud detection using rule-based logic

- Search and filter functionality
- Admin control for modifying transaction status
- User-specific transaction visibility
- Clean and responsive UI

5. Technologies Used

- **Frontend:** HTML, CSS, Angular
- **Programming Language:** TypeScript
- **Logic Implementation:** JavaScript concepts
- **Data Handling:** LocalStorage
- **Architecture:** Component and Service-based Angular structure

6. System Workflow

Login → Role Identification → Dashboard Load
 → Fetch Transactions → Apply Fraud Rules
 → Display Results (Fraud / Normal)

7. Expected Outcome

The system provides an efficient way to monitor banking transactions and detect fraud in real-time. It reduces manual effort, improves security, and enhances transparency for both users and administrators.

The application ensures that users can securely view their data while administrators have full control over monitoring and managing transactions.

8. Conclusion

The Banking Transaction Monitoring System is an effective solution for modern banking needs. It simplifies transaction monitoring through automation and rule-based fraud detection.

With its user-friendly interface and efficient design, the system enhances financial security and provides better control over transaction management, making it suitable for real-world banking applications.